Lab 3 - Routing Protocol

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SECTION: 01

Introduction

You are given a Packet Tracer file, which requires some work IP addressing and routing protocol configuration. You must follow all the steps carefully and answer the given questions.

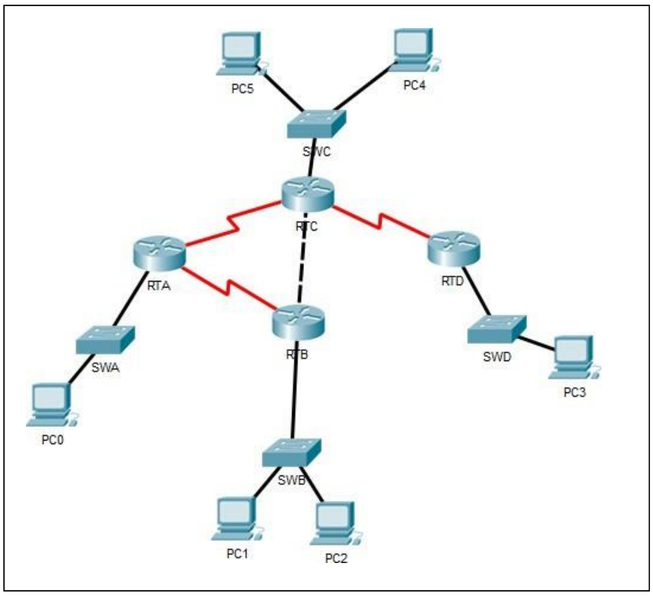


Figure 1

Task 1: IP addressing

Step 1: Fill in Table 1 below with the correct information. Note: The information may be found under the ***Config*** tab of each router (refer to Figure2).

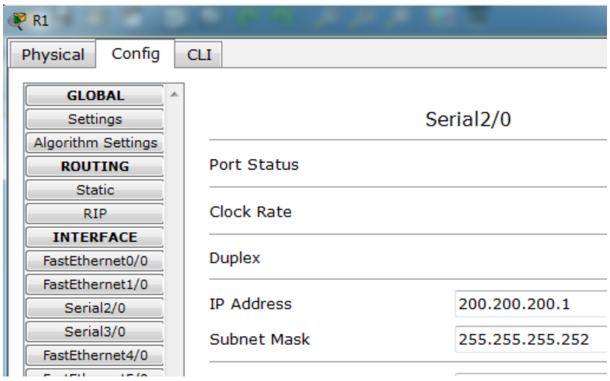


Figure 2

Table 1

#	Device Name	Interface	IP Address	Subnet Mask
1	RTA	Se2/0	172.16.230.5	255.255.255.252
2		Se3/0	172.16.230.1	255.255.255.252
3		Fa0/0	172.16.224.255	255.255.254.0
4	RTB	Se2/0	172.16.230.2	255.255.255.252
5		Fa0/0	172.16.230.9	255.255.255.252
6		Fa1/0	172.16.226.11	255.255.254.0
7	RTC	Se2/0	172.16.230.6	255.255.255.252
8		Se3/0	172.16.230.13	255.255.255.252
9		Fa0/0	172.16.230.10	255.255.255.252
10		Fa1/0	172.16.228.11	255.255.255.0
10	RTD	Se2/0	172.16.230.14	255.255.255.252
11		Fa0/0	172.16.229.222	255.255.255.0

Step 2: Given the information in file, answer the following questions:

a. How many different subnets are there? 8

b. What is the network address of each of these subnets? (*Hint: Given the IP address and the subnet mask, you can calculate the network address using AND operation*). Label the subnets in the topology given in Figure 1, and complete Table 2 below.

Table 2

Subnet #	Network Address	Broadcast Address	Range of usable addresses
1	172.16.224.0	172.16.225.255	172.16.224.1 - 172.16.225.254
2	172.16.230.0	172.16.230.3	172.16.230.1 - 172.16.230.2
3	172.16.230.4	172.16.230.7	172.16.230.5 - 172.16.230.6
4	172.16.226.0	172.16.227.255	172.16.226.1 - 172.16.227.254
5	172.16.230.8	172.16.230.11	172.16.230.9 - 172.16.230.10
6	172.16.228.0	172.16.228.255	172.16.228.1 - 172.16.228.254
7	172.16.229.0	172.16.229.255	172.16.229.1 - 172.16.229.254
8	172.16.230.12	172.16.230.15	172.16.230.13 - 172.16.230.14

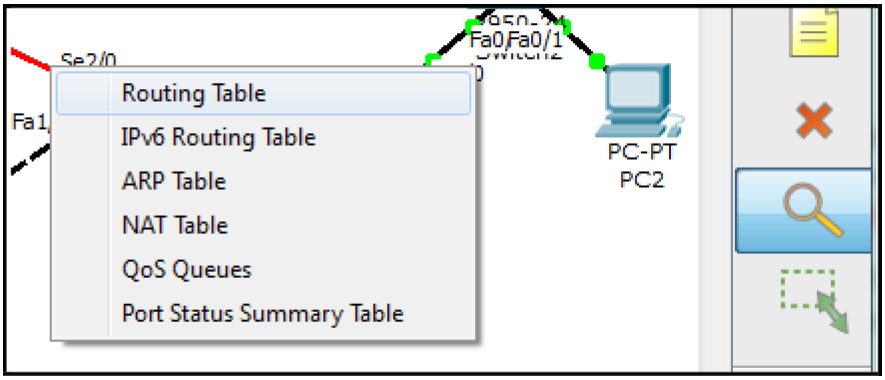
c. Provided that all PC will use the last usable address in its subnet, fill in Table 3 below with the correct information.

Table 3

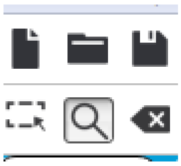
#	Device Name	IP Address	Subnet Mask	Default Gateway
1	PCA	172.16.225.254	255.255.254.0	172.16.224.255
2	PCB	172.16.227.254	255.255.254.0	172.16.226.11
3	PCC	172.16.228.254	255.255.255.0	172.16.228.11
4	PCD	172.16.229.254	255.255.255.0	172.16.229.222

Step 3: Complete the IP addressing information on all the PCs in the topology. (Hint: Click on the PC, choose the ***Desktop*** tab, then click ***IP Configuration***).

Step 4: Open the routing table for each router. *(Hint: you can use the 'magnifying glass' icon, then point to a router and choose 'Routing Table'. See Figure 3 below.)*



(a)



(b)

Figure 3

Step 5: Copy the image of the routing table for each router. *(Hint: You can use 'Snipping Tool' to copy the image.)*

1. RTA

Routing Table for RTA

Type	Network	Port	Next Hop IP	Metric
C	172.16.224.0/23	FastEthernet0/0	---	0/0
C	172.16.230.0/30	Serial3/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0

2. RTB

Routing Table for RTB

Type	Network	Port	Next Hop IP	Metric
C	172.16.226.0/23	FastEthernet1/0	---	0/0
C	172.16.230.0/30	Serial2/0	---	0/0
C	172.16.230.8/30	FastEthernet0/0	---	0/0

3. RTC

Routing Table for RTC

Type	Network	Port	Next Hop IP	Metric
C	172.16.228.0/24	FastEthernet1/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0
C	172.16.230.8/30	FastEthernet0/0	---	0/0
C	172.16.230.12/30	Serial3/0	---	0/0

4. RTD

Routing Table for RTD

Type	Network	Port	Next Hop IP	Metric
C	172.16.229.0/24	FastEthernet0/0	---	0/0
C	172.16.230.12/30	Serial2/0	---	0/0

Step 6: Answer the questions below.

a. Do all the routers have the same information in its routing table?
No.

b. What is the difference that can be seen?
According to routing table of each router, it is obvious that the routers only have directly connected networks since the type of all routers is 'C' which identified a directly connected network.

c. Can all the PCs ping each other successfully?

According to the Table 4 below, all the PCs do not ping each other successfully.

Table 4

#	**Ping between devices**	**Successful (✓)**	**Unsuccessful (×)**
1	PCA-PCB		×
2	PCA-PCC		×
3	PCA-PCD		×
4	PCB-PCC		×
5	PCB-PCD		×
6	PCC-PCD		×

d. Reflection: what is the reason for your answer in (c)?

This is because there is the PCs do not have path to other networks which cause the router does not have any information to forward the packet.

Task 2: Dynamic routing configuration – RIP

Dynamic routing allows the network to be more flexible to changes. It can help the routers adapt to the changes in the pathways without much intervention from network administrators.

In this part of the lab, you will learn how to configure RIP routing protocol, and see how changes happen in the routing tables. Routers R1 and R4 is already configured for you.

Step 1: Choose Router RTA. Click the CLI tab. Copy the following text into the command line interface.

RTA>***enable***
RTA#***configure terminal***
Enter configuration commands, one per line. End with CNTL/Z.
RTA(config)#***router rip***
RTA(config-router)#***version 2***
RTA(config-router)#***network 172.16.0.0***
RTA(config-router)#***no auto-summary***
RTA(config-router)#***exit***
RTA(config)#***exit***
RTA#
%SYS-5-CONFIG_I: Configured from console by console
RTA#***copy running-config startup-config***
Destination filename [startup-config]?
Building configuration...
[OK]
RTA#

When asked this just press ENTER

Task 1.1:

(a) Copy (paste image) of the RTA routing table here.

Routing Table for RTA

Type	Network	Port	Next Hop IP	Metric
C	172.16.224.0/23	FastEthernet0/0	---	0/0
C	172.16.230.0/30	Serial3/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0

Step 2: Choose Router RTB. Click the CLI tab. Copy the following text into the command line interface.

RTB>***enable***
RTB#***configure terminal***
Enter configuration commands, one per line. End with CNTL/Z.
RTB(config)#***router rip***
RTB(config-router)#***version 2***
RTB(config-router)#***network 172.16.0.0***
RTB(config-router)#***no auto-summary***
RTB(config-router)#***exit***
RTB(config)#***exit***
RTB#
%SYS-5-CONFIG_I: Configured from console by console
RTB#***copy running-config startup-config***
Destination filename [startup-config]?
Building configuration...
[OK]
RTB#

Task 2.1:

(a) Copy (paste image) of the RTB routing table and a new look into RTA routing table here.
RTB:

Routing Table for RTB

Type	Network	Port	Next Hop IP	Metric
R	172.16.224.0/23	Serial2/0	172.16.230.1	120/1
C	172.16.226.0/23	FastEthernet1/0	---	0/0
C	172.16.230.0/30	Serial2/0	---	0/0
R	172.16.230.4/30	Serial2/0	172.16.230.1	120/1
C	172.16.230.8/30	FastEthernet0/0	---	0/0

RTA:

Routing Table for RTA

Type	Network	Port	Next Hop IP	Metric
C	172.16.224.0/23	FastEthernet0/0	---	0/0
R	172.16.226.0/23	Serial3/0	172.16.230.2	120/1
C	172.16.230.0/30	Serial3/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0
R	172.16.230.8/30	Serial3/0	172.16.230.2	120/1

(b) **Reflection:** what difference do you see between routing tables of RTA and RTB?
According to the routing tables of RTA and RTB, we can determine the 'Next Hop IP' in the tables which show the routers are connected to each other. Therefore, we can know RTA and RTB are exchanging the route information of each other.

Step 3: Copy the same configuration instructions to RTC and RTD

Step 4: Answer the questions below.

a. Do all the routers have the same information in its routing table?
No. All the routers have the different information in its routing table.

RTA:

Routing Table for RTA

Type	Network	Port	Next Hop IP	Metric
C	172.16.224.0/23	FastEthernet0/0	---	0/0
R	172.16.226.0/23	Serial3/0	172.16.230.2	120/1
R	172.16.228.0/24	Serial2/0	172.16.230.6	120/1
R	172.16.229.0/24	Serial2/0	172.16.230.6	120/2
C	172.16.230.0/30	Serial3/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0
R	172.16.230.8/30	Serial3/0	172.16.230.2	120/1
R	172.16.230.8/30	Serial2/0	172.16.230.6	120/1
R	172.16.230.12/30	Serial2/0	172.16.230.6	120/1

RTB:

Routing Table for RTB

Type	Network	Port	Next Hop IP	Metric
R	172.16.224.0/23	Serial2/0	172.16.230.1	120/1
C	172.16.226.0/23	FastEthernet1/0	---	0/0
R	172.16.228.0/24	FastEthernet0/0	172.16.230.10	120/1
R	172.16.229.0/24	FastEthernet0/0	172.16.230.10	120/2
C	172.16.230.0/30	Serial2/0	---	0/0
R	172.16.230.4/30	Serial2/0	172.16.230.1	120/1
R	172.16.230.4/30	FastEthernet0/0	172.16.230.10	120/1
C	172.16.230.8/30	FastEthernet0/0	---	0/0
R	172.16.230.12/30	FastEthernet0/0	172.16.230.10	120/1

RTC:

Routing Table for RTC

Type	Network	Port	Next Hop IP	Metric
R	172.16.224.0/23	Serial2/0	172.16.230.5	120/1
R	172.16.226.0/23	FastEthernet0/0	172.16.230.9	120/1
C	172.16.228.0/24	FastEthernet1/0	---	0/0
R	172.16.229.0/24	Serial3/0	172.16.230.14	120/1
R	172.16.230.0/30	Serial2/0	172.16.230.5	120/1
R	172.16.230.0/30	FastEthernet0/0	172.16.230.9	120/1
C	172.16.230.4/30	Serial2/0	---	0/0
C	172.16.230.8/30	FastEthernet0/0	---	0/0
C	172.16.230.12/30	Serial3/0	---	0/0

RTD:

Routing Table for RTD

Type	Network	Port	Next Hop IP	Metric
R	172.16.224.0/23	Serial2/0	172.16.230.13	120/2
R	172.16.226.0/23	Serial2/0	172.16.230.13	120/2
R	172.16.228.0/24	Serial2/0	172.16.230.13	120/1
C	172.16.229.0/24	FastEthernet0/0	---	0/0
R	172.16.230.0/30	Serial2/0	172.16.230.13	120/2
R	172.16.230.4/30	Serial2/0	172.16.230.13	120/1
R	172.16.230.8/30	Serial2/0	172.16.230.13	120/1
C	172.16.230.12/30	Serial2/0	---	0/0

b. Write down what RTC and RTD routing table information (Next Hop IP, Metric) to the network 172.16.224.0/24.

	Next Hop IP	Metric
RTC	**172.16.230.5**	**120/1**
RTD	**172.16.230.13**	**120/2**

c. What is the difference that can be seen? Why is this?
The difference that can be seen is RTC and RTD have different Next Hop IP because they have different next gateway that the packet goes through. Other than that, RTD also have bigger metric which is 120/2 because RTD need to go through RTC to get to RTA which is the best path but it is further compared to RTC to RTA, therefore, RTD has bigger metric.

d. Can all the PCs ping each other successfully?

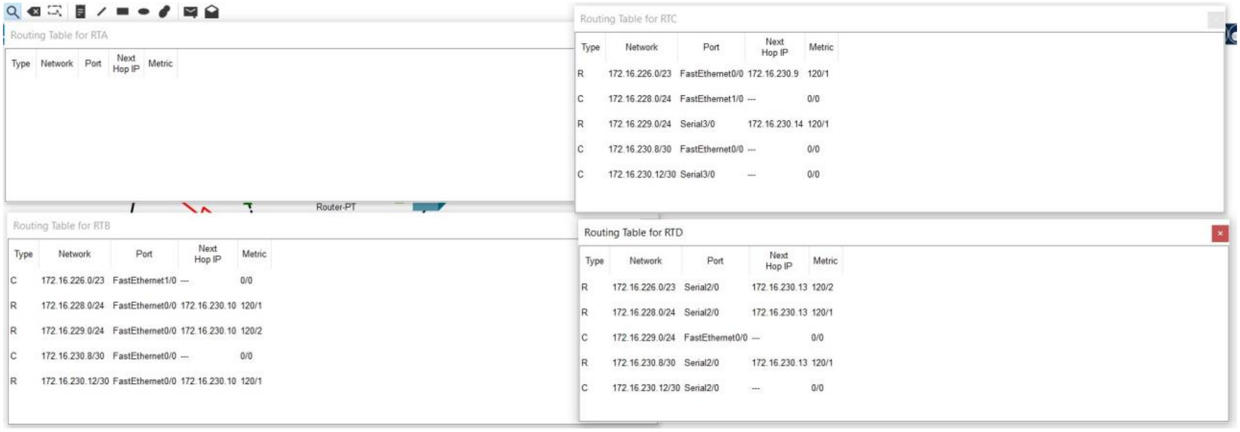
Table 5

#	**Ping between devices**	**Successful (✓)**	**Unsuccessful (×)**
1	PCA-PCB	✓	
2	PCA-PCC	✓	
3	PCA-PCD	✓	
4	PCB-PCC	✓	
5	PCB-PCD	✓	
6	PCC-PCD	✓	

e. Reflection: what is the reason for your answer in (d)?
All the routers can start to exchange the information since we have configured and built the routing table. Hence, the packets can be forward to the destinations successfully.

Step 5: Switch off router RTA. What are the changes noted in the routing tables?

The routing table of router RTA has no information, and the 172.16.224.0/23 is disappeared from the routing tables of RTB, RTC and RTD.



Routing Table for RTA

Type	Network	Port	Next Hop IP	Metric

Routing Table for RTC

Type	Network	Port	Next Hop IP	Metric
R	172.16.226.0/23	FastEthernet0/0	172.16.230.9	120/1
C	172.16.228.0/24	FastEthernet1/0	---	0/0
R	172.16.229.0/24	Serial3/0	172.16.230.14	120/1
C	172.16.230.8/30	FastEthernet0/0	---	0/0
C	172.16.230.12/30	Serial3/0	---	0/0

Routing Table for RTB

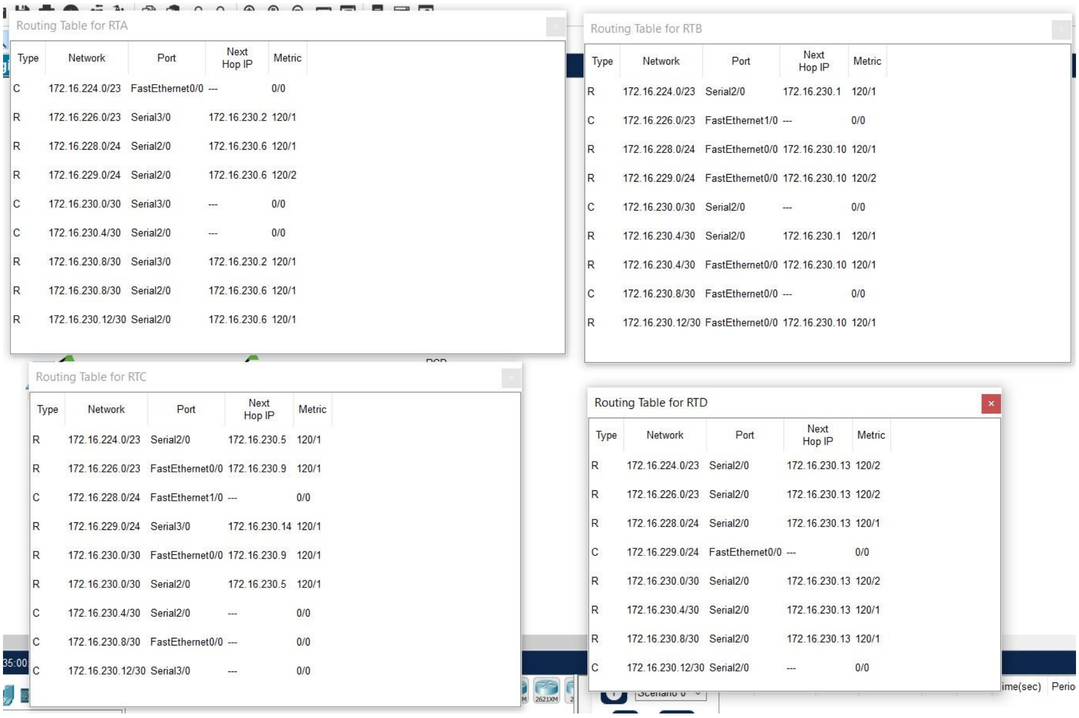
Type	Network	Port	Next Hop IP	Metric
C	172.16.226.0/23	FastEthernet1/0	---	0/0
R	172.16.228.0/24	FastEthernet0/0	172.16.230.10	120/1
R	172.16.229.0/24	FastEthernet0/0	172.16.230.10	120/2
C	172.16.230.8/30	FastEthernet0/0	---	0/0
R	172.16.230.12/30	FastEthernet0/0	172.16.230.10	120/1

Routing Table for RTD

Type	Network	Port	Next Hop IP	Metric
R	172.16.226.0/23	Serial2/0	172.16.230.13	120/2
R	172.16.228.0/24	Serial2/0	172.16.230.13	120/1
C	172.16.229.0/24	FastEthernet0/0	---	0/0
R	172.16.230.8/30	Serial2/0	172.16.230.13	120/1
C	172.16.230.12/30	Serial2/0	---	0/0

Step 6: Switch on router RTA. What are the changes noted in the routing tables?

The routing table of router RTA have all the information back and the 172.16.224.0/23 disappeared before also appear back in the routing tables of RTB, RTC and RTD.



Routing Table for RTA

Type	Network	Port	Next Hop IP	Metric
C	172.16.224.0/23	FastEthernet0/0	---	0/0
R	172.16.226.0/23	Serial3/0	172.16.230.2	120/1
R	172.16.228.0/24	Serial2/0	172.16.230.6	120/1
R	172.16.229.0/24	Serial2/0	172.16.230.6	120/2
C	172.16.230.0/30	Serial3/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0
R	172.16.230.8/30	Serial3/0	172.16.230.2	120/1
R	172.16.230.8/30	Serial2/0	172.16.230.6	120/1
R	172.16.230.12/30	Serial2/0	172.16.230.6	120/1

Routing Table for RTB

Type	Network	Port	Next Hop IP	Metric
R	172.16.224.0/23	Serial2/0	172.16.230.1	120/1
C	172.16.226.0/23	FastEthernet1/0	---	0/0
R	172.16.228.0/24	FastEthernet0/0	172.16.230.10	120/1
R	172.16.229.0/24	FastEthernet0/0	172.16.230.10	120/2
C	172.16.230.0/30	Serial2/0	---	0/0
R	172.16.230.4/30	Serial2/0	172.16.230.1	120/1
R	172.16.230.4/30	FastEthernet0/0	172.16.230.10	120/1
C	172.16.230.8/30	FastEthernet0/0	---	0/0
R	172.16.230.12/30	FastEthernet0/0	172.16.230.10	120/1

Routing Table for RTC

Type	Network	Port	Next Hop IP	Metric
R	172.16.224.0/23	Serial2/0	172.16.230.5	120/1
R	172.16.226.0/23	FastEthernet0/0	172.16.230.9	120/1
C	172.16.228.0/24	FastEthernet1/0	---	0/0
R	172.16.229.0/24	Serial3/0	172.16.230.14	120/1
R	172.16.230.0/30	FastEthernet0/0	172.16.230.9	120/1
R	172.16.230.0/30	Serial2/0	172.16.230.5	120/1
C	172.16.230.4/30	Serial2/0	---	0/0
C	172.16.230.8/30	FastEthernet0/0	---	0/0
C	172.16.230.12/30	Serial3/0	---	0/0

Routing Table for RTD

Type	Network	Port	Next Hop IP	Metric
R	172.16.224.0/23	Serial2/0	172.16.230.13	120/2
R	172.16.226.0/23	Serial2/0	172.16.230.13	120/2
R	172.16.228.0/24	Serial2/0	172.16.230.13	120/1
C	172.16.229.0/24	FastEthernet0/0	---	0/0
R	172.16.230.0/30	Serial2/0	172.16.230.13	120/2
R	172.16.230.4/30	Serial2/0	172.16.230.13	120/1
R	172.16.230.8/30	Serial2/0	172.16.230.13	120/1
C	172.16.230.12/30	Serial2/0	---	0/0

Step 7: Reflection: What have you learned in this task?

Through setting up the Router Information Protocol, the routers will fill up their routing tables by listening to the next router. When there is any router being removed or added to the network, the routing table of the router will be updated.

Task 3: Dynamic routing configuration – OSPF

Make sure that you have all the routing tables on display on one side (as before). As you go through the steps, look at the changes happening in the routing tables.

Step 1: On **all** the routers, do the following.

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no router rip
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#copy running-config startup-config
Destination filename [startup-config]?
```

Step 2: Copy the image of the routing table for each router.

RTA:

Routing Table for RTA

Type	Network	Port	Next Hop IP	Metric
C	172.16.224.0/23	FastEthernet0/0	---	0/0
C	172.16.230.0/30	Serial3/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0

RTB:

Routing Table for RTB

Type	Network	Port	Next Hop IP	Metric
C	172.16.226.0/23	FastEthernet1/0	---	0/0
C	172.16.230.0/30	Serial2/0	---	0/0
C	172.16.230.8/30	FastEthernet0/0	---	0/0

RTC:

Routing Table for RTC

Type	Network	Port	Next Hop IP	Metric
C	172.16.228.0/24	FastEthernet1/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0
C	172.16.230.8/30	FastEthernet0/0	---	0/0
C	172.16.230.12/30	Serial3/0	---	0/0

RTD:

Routing Table for RTD

Type	Network	Port	Next Hop IP	Metric
C	172.16.229.0/24	FastEthernet0/0	---	0/0
C	172.16.230.12/30	Serial2/0	---	0/0

Step 3: For Router RTA, Click the CLI tab. Copy the following text into the command line interface.

RTA# ***configure terminal***
Enter configuration commands, one per line. End with CNTL/Z.
RTA(config)#***router ospf 1***
RTA(config-router)#***network 172.16.224.0 0.0.1.255 area 0***
RTA(config-router)#***network 172.16.230.0 0.0.0.3 area 0***
RTA(config-router)#***network 172.16.230.4 0.0.0.3 area 0***
RTA(config-router)#***end***
RTA# ***copy running-config startup-config***
Destination filename [startup-config]?
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
RTA#

Task 3.1: paste the image of RTA's routing table here.

Routing Table for RTA

Type	Network	Port	Next Hop IP	Metric
C	172.16.224.0/23	FastEthernet0/0	---	0/0
C	172.16.230.0/30	Serial3/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0

Task 3.2:

a. Does RTA have a path to ALL the different subnet?
   No. RTA does not have a path to ALL the different subnet.

b. Try pinging the different PCs and jot down your results.

Table

#	Ping between devices	Successful (✓)	Unsuccessful (×)
1	PCA-PCB		×
2	PCA-PCC		×
3	PCA-PCD		×

Step 4: Configure the other routers with OSPF routing algorithm.

Step 4.1: For Router RTB, Click the CLI tab. Copy the following text into the command line interface.

RTB# ***configure terminal***
Enter configuration commands, one per line. End with CNTL/Z.
RTB(config)#***router ospf 1***
RTB(config-router)#***network 172.16.226.0 0.0.1.255 area 0***
RTB(config-router)#***network 172.16.230.0 0.0.0.3 area 0***
RTB(config-router)#***network 172.16.230.8 0.0.0.3 area 0***
RTB(config-router)#end
RTB# ***copy running-config startup-config***
Destination filename [startup-config]?
%SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
RTB#

Step 4.2: For Router RTC, Click the CLI tab. Copy the following text into the command line interface.

RTC# ***configure terminal***
Enter configuration commands, one per line. End with CNTL/Z.
RTC(config)#***router ospf 1***
RTC(config-router)#***network 172.16.228.0 0.0.0.255 area 0***
RTC(config-router)#***network 172.16.230.4 0.0.0.3 area 0***
RTC(config-router)#***network 172.16.230.8 0.0.0.3 area 0***
RTC(config-router)#***network 172.16.230.12 0.0.0.3 area 0***
RTC(config-router)#***end***
RTC# ***copy running-config startup-config***
Destination filename [startup-config]?
%SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
RTC#

Step 4.3: For Router RTD, Click the CLI tab. Copy the following text into the command line interface.

RTD# ***configure terminal***
Enter configuration commands, one per line. End with CNTL/Z.
RTD(config)#
RTD(config)#***router ospf 1***
RTD(config-router)#***network 172.16.229.0 0.0.0.255 area 0***
RTD(config-router)#***network 172.16.230.12 0.0.0.3 area 0***
RTD(config-router)#***end***
RTD# ***copy running-config startup-config***
Destination filename [startup-config]?
Building configuration...
[OK]
RTD#
%SYS-5-CONFIG_I: Configured from console by console
RTD#

Step 5: Copy the image of the routing table for each router and paste it here.
RTA:

Routing Table for RTA

Type	Network	Port	Next Hop IP	Metric
C	172.16.224.0/23	FastEthernet0/0	---	0/0
O	172.16.226.0/23	Serial3/0	172.16.230.2	110/65
O	172.16.228.0/24	Serial2/0	172.16.230.6	110/65
O	172.16.229.0/24	Serial2/0	172.16.230.6	110/129
C	172.16.230.0/30	Serial3/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0
O	172.16.230.8/30	Serial3/0	172.16.230.2	110/65
O	172.16.230.8/30	Serial2/0	172.16.230.6	110/65
O	172.16.230.12/30	Serial2/0	172.16.230.6	110/128

RTB:

Routing Table for RTB

Type	Network	Port	Next Hop IP	Metric
O	172.16.224.0/23	Serial2/0	172.16.230.1	110/65
C	172.16.226.0/23	FastEthernet1/0	---	0/0
O	172.16.228.0/24	FastEthernet0/0	172.16.230.10	110/2
O	172.16.229.0/24	FastEthernet0/0	172.16.230.10	110/66
C	172.16.230.0/30	Serial2/0	---	0/0
O	172.16.230.4/30	FastEthernet0/0	172.16.230.10	110/65
C	172.16.230.8/30	FastEthernet0/0	---	0/0
O	172.16.230.12/30	FastEthernet0/0	172.16.230.10	110/65

RTC:

Routing Table for RTC

Type	Network	Port	Next Hop IP	Metric
O	172.16.224.0/23	Serial2/0	172.16.230.5	110/65
O	172.16.226.0/23	FastEthernet0/0	172.16.230.9	110/2
C	172.16.228.0/24	FastEthernet1/0	---	0/0
O	172.16.229.0/24	Serial3/0	172.16.230.14	110/65
O	172.16.230.0/30	FastEthernet0/0	172.16.230.9	110/65
C	172.16.230.4/30	Serial2/0	---	0/0
C	172.16.230.8/30	FastEthernet0/0	---	0/0
C	172.16.230.12/30	Serial3/0	---	0/0

RTD:

Routing Table for RTD

Type	Network	Port	Next Hop IP	Metric
O	172.16.224.0/23	Serial2/0	172.16.230.13	110/129
O	172.16.226.0/23	Serial2/0	172.16.230.13	110/66
O	172.16.228.0/24	Serial2/0	172.16.230.13	110/65
C	172.16.229.0/24	FastEthernet0/0	---	0/0
O	172.16.230.0/30	Serial2/0	172.16.230.13	110/129
O	172.16.230.4/30	Serial2/0	172.16.230.13	110/128
O	172.16.230.8/30	Serial2/0	172.16.230.13	110/65
C	172.16.230.12/30	Serial2/0	---	0/0

Step 6: Switch off router RTA. What are the changes noted in the routing tables?

The routing table of router RTA has no information, and the 172.16.224.0/23 is disappeared from the routing tables of RTB, RTC and RTD.

Routing Table for RTA

Type	Network	Port	Next Hop IP	Metric

Routing Table for RTC

Type	Network	Port	Next Hop IP	Metric
O	172.16.226.0/23	FastEthernet0/0	172.16.230.9	110/2
C	172.16.228.0/24	FastEthernet1/0	---	0/0
O	172.16.229.0/24	Serial3/0	172.16.230.14	110/65
C	172.16.230.8/30	FastEthernet0/0	---	0/0
C	172.16.230.12/30	Serial3/0	---	0/0

Routing Table for RTB

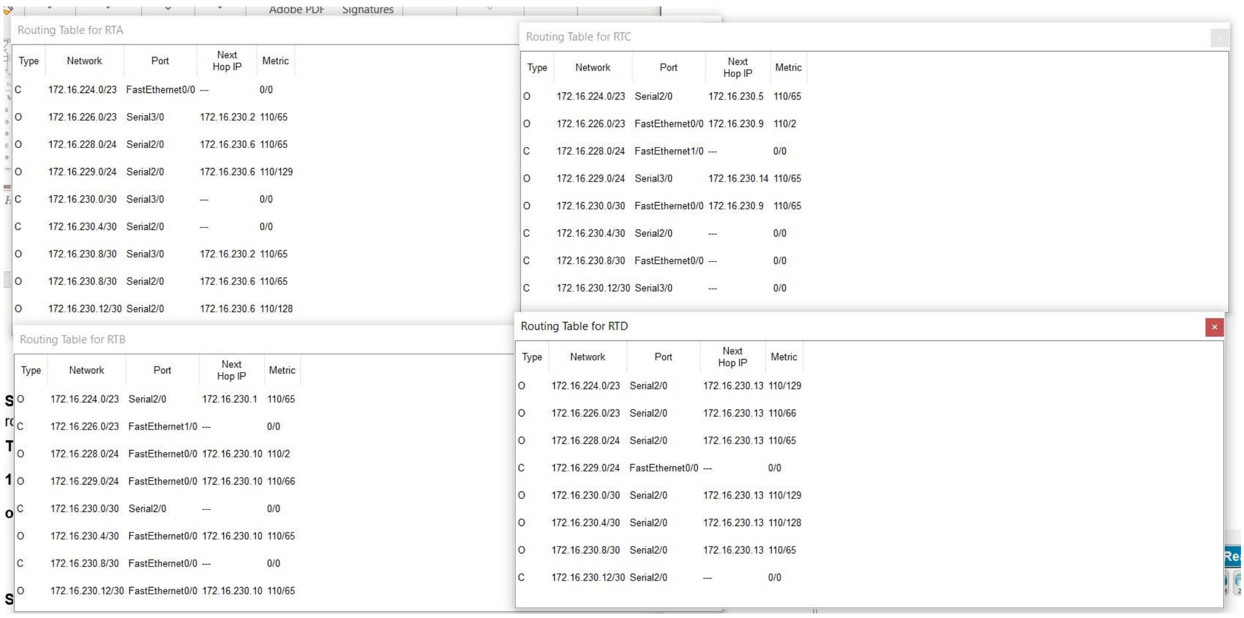
Type	Network	Port	Next Hop IP	Metric
C	172.16.226.0/23	FastEthernet1/0	---	0/0
O	172.16.228.0/24	FastEthernet0/0	172.16.230.10	110/2
O	172.16.229.0/24	FastEthernet0/0	172.16.230.10	110/66
C	172.16.230.8/30	FastEthernet0/0	---	0/0
O	172.16.230.12/30	FastEthernet0/0	172.16.230.10	110/65

Routing Table for RTD

Type	Network	Port	Next Hop IP	Metric
O	172.16.226.0/23	Serial2/0	172.16.230.13	110/66
O	172.16.228.0/24	Serial2/0	172.16.230.13	110/65
C	172.16.229.0/24	FastEthernet0/0	---	0/0
O	172.16.230.8/30	Serial2/0	172.16.230.13	110/65
C	172.16.230.12/30	Serial2/0	---	0/0

Step 7: Switch on router RTA. Wait a few minutes. What are the changes noted in the routing tables?

The routing table of router RTA have all the information back and the 172.16.224.0/23 disappeared before also appear back in the routing tables of RTB, RTC and RTD.



Routing Table for RTA

Type	Network	Port	Next Hop IP	Metric
C	172.16.224.0/23	FastEthernet0/0	---	0/0
O	172.16.226.0/23	Serial3/0	172.16.230.2	110/65
O	172.16.228.0/24	Serial2/0	172.16.230.6	110/65
O	172.16.229.0/24	Serial2/0	172.16.230.6	110/129
C	172.16.230.0/30	Serial3/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0
O	172.16.230.8/30	Serial3/0	172.16.230.2	110/65
O	172.16.230.8/30	Serial2/0	172.16.230.6	110/65
O	172.16.230.12/30	Serial2/0	172.16.230.6	110/128

Routing Table for RTC

Type	Network	Port	Next Hop IP	Metric
O	172.16.224.0/23	Serial2/0	172.16.230.5	110/65
O	172.16.226.0/23	FastEthernet0/0	172.16.230.9	110/2
C	172.16.228.0/24	FastEthernet1/0	---	0/0
O	172.16.229.0/24	Serial3/0	172.16.230.14	110/65
O	172.16.230.0/30	FastEthernet0/0	172.16.230.9	110/65
C	172.16.230.4/30	Serial2/0	---	0/0
C	172.16.230.8/30	FastEthernet0/0	---	0/0
C	172.16.230.12/30	Serial3/0	---	0/0

Routing Table for RTB

Type	Network	Port	Next Hop IP	Metric
O	172.16.224.0/23	Serial2/0	172.16.230.1	110/65
C	172.16.226.0/23	FastEthernet1/0	---	0/0
O	172.16.228.0/24	FastEthernet0/0	172.16.230.10	110/2
O	172.16.229.0/24	FastEthernet0/0	172.16.230.10	110/66
C	172.16.230.0/30	Serial2/0	---	0/0
O	172.16.230.4/30	FastEthernet0/0	172.16.230.10	110/65
C	172.16.230.8/30	FastEthernet0/0	---	0/0
O	172.16.230.12/30	FastEthernet0/0	172.16.230.10	110/65

Routing Table for RTD

Type	Network	Port	Next Hop IP	Metric
O	172.16.224.0/23	Serial2/0	172.16.230.13	110/129
O	172.16.226.0/23	Serial2/0	172.16.230.13	110/66
O	172.16.228.0/24	Serial2/0	172.16.230.13	110/65
C	172.16.229.0/24	FastEthernet0/0	---	0/0
O	172.16.230.0/30	Serial2/0	172.16.230.13	110/129
O	172.16.230.4/30	Serial2/0	172.16.230.13	110/128
O	172.16.230.8/30	Serial2/0	172.16.230.13	110/65
C	172.16.230.12/30	Serial2/0	---	0/0

Step 8: Reflection: What have you learned in this task?

As what I learned from this task, the Routing Information Protocol (RIP) is a protocol that applicable for small network as it delivers the whole routing table to all the routers in the network every 30 seconds. On the other hand, Open Shortest Path First (OSPF) is a protocol that able to bring back the connection normally without making configuration again when one router is switched off. Moreover, OSPF only exchange routing information when there is a change in the network topology.

.....END.......